

Key: Lines Drawn in 1937

84-355 Democracy's Data | Redlining | Instructor copy — do not distribute

Jonathan Cervas

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All code is given to students. Grade the interpretation.

Structure, as of the rebuild. The brief (`redlining-brief.Rmd`) is now a chapter, “Hazardous”: it is organised by argument under prose headings and carries four numbered D3 figures rather than numbered steps — Figure 1, Cleveland as the HOLC graded it in 1939; Figure 2, a window of that city with 2020 tracts laid over the 1939 areas; Figure 3, how much of each tract is really one grade; Figure 4, the within-city A-to-D gap for every city. The lab (`redlining.Rmd`) is unchanged and still runs Steps 1–10 with the code exposed. **Students should read the brief first. Step numbers below refer to the lab.** The chapter covers the same ground in the same order, so each block of lab Steps has a heading facing it: Steps 1–3 = “Best, Still Desirable, Definitely Declining, Hazardous” and “Two geographies that were never meant to meet” (Figures 1 and 2), with “The cost of the rule, measured” (Figure 3); Steps 4–5 = “The answer that arrives first is the wrong one”; Steps 6–7 = “Holding the city fixed” (Figure 4); Steps 8–9 = “The cities that run the other way” and “Did the maps do anything, or only record what was already there?”; Step 10 = “What the ground can testify to”.

The maps are now in the documents. The chapter opens on the appraisal sheet for area D9 quoted verbatim — the *Clarifying Remarks*, the *Infiltration of line*, and the appraiser’s own count of 0% Black residents — and Figure 1 then draws the real 1939 HOLC map of Cleveland in the HOLC’s own four colours. Figure 2 lays the 2020 tract boundaries over the same polygons and shows the rule failing on a named tract. The lab draws the Cleveland map too. All of it comes from Mapping Inequality’s digitised polygons via `data/build-brief-figures.R`. **You no longer have to pull the maps up on a projector — but do it anyway if the room is engaged, and read a second area description aloud.**

One of three labs in the discrimination unit, alongside policing and jury selection. Run in parallel, then compare.

This one is different from the other two in a way worth naming out loud. There, the discriminatory intent is unrecorded and has to be inferred. Here the intent is written on the document — HOLC appraisers named Black residents as a reason for a low grade. What has to be shown is not intent but **persistence**, and that turns out to be the harder problem.

Every number below was run against the data files.

The session in one sentence

The comparison that gets published says the 1937 maps barely matter; doing it one city at a time says they matter enormously; and the cities where it reverses explain why both are true.

The findings

```
round(setNames(g$pct, g$grade), 1)
```

```
#>      A      B      C      D
#> 16.7 23.3 21.4 25.6
```

```
c(pooled_gap_D_minus_A = round(g$pct[4] - g$pct[1], 1),
  mean_within_city_gap = round(mean(ci$gap), 1),
  cities = nrow(ci),
  cities_reversed = sum(ci$gap < 0))
```

```
#> pooled_gap_D_minus_A mean_within_city_gap      cities
#>                8.9                14.5                33.0
#>      cities_reversed
#>                6.0
```

- Pooled: D-graded tracts **25.6%** Black, A-graded **16.7%** — a gap of 8.9 points, ratio 1.5.
- **Mean within-city gap: 14.5 points.** Median 11.4.
- Cleveland **+46.9**, Chicago **+33.4**, Pittsburgh **+29.9**.
- **6 of 33 cities run backwards**, Philadelphia among them.

The three things to get said

1. Step 4 is designed to disappoint, and students must feel that. A ratio of 1.5 looks like weak evidence for a famous claim, and some will be ready to conclude the redlining story is overstated. **Let them sit with it for a minute.** The reveal in Step 6 only works if Step 4 was believed.

2. The pooling problem is the transferable lesson. Adding cities together compares a D tract in Chicago against an A tract in Los Angeles — cities with different racial compositions for reasons unrelated to HOLC grades. **This is the aggregation problem the course has met three times** (EAVS jurisdictions, precinct units, ecological inference), arriving with the largest consequence: the pooled estimate is barely half the within-city one.

3. The reversals are evidence, not noise. Six cities run backwards, and the explanation is mid-century white flight — the “Best” 1937 neighbourhoods in Philadelphia and Detroit are exactly the ones that later became majority Black. **A student who wants to drop those cities is doing the wrong thing**, and saying so is the most useful thirty seconds in the session.

The line for the board: an average over places is not a finding about any place.

Handling the room

Less volatile than the other two labs — the intent is documented, so there is nothing to argue about on that front. Two risks instead:

- **Overclaiming causation.** The maps recorded existing segregation as well as entrenching it. The lab says so; enforce it. “Redlining caused this” is not supported and is not necessary.
- **Fatalism.** Ninety-year-old lines still visible can read as immovable. The reversals are the antidote — in some cities the pattern did not persist, and the reason was that everything else changed.



Walking the steps

Steps	What they do	Min	Watch for
1-3	Two sources 80 years apart; the join that has to be built; one record	10	Point-in-polygon is an assumption. Say so before any result. The close-up in the brief, Figure 2, shows it failing; use it. 16.7% vs 25.6% is deliberately underwhelming. Let it disappoint.
4-5	The pooled comparison, and what is wrong with it	10	The pivot. Cleveland: 21.3 vs 68.1 in one housing market. Philadelphia is gentrification, not absence of effect. Small-denominator cities are noise.
6-7	Within cities; then every city at once	12	Association within cities, yes. Causation, not from this design.
8-9	The six reversals, and what they mean for the claim	10	
10	What you can say — and what needs Aaronson et al.	6	

Steps 4-6 are the pivot. The pooled national comparison must be believed for a minute before it is broken. The lesson is the unit of comparison, not the number.

If you are short on time, protect the pivot and the closing step. The numbers are what make the argument concrete; the sequence is the argument.

Option A — Weight by population

Verified output

```
c(pooled_gap      = round(g$pct[4] - g$pct[1], 1),
  mean_of_city_gaps = round(mean(ci$gap), 1),
  median_city_gap   = round(median(ci$gap), 1))
```

```
#>      pooled_gap mean_of_city_gaps  median_city_gap
#>           8.9           14.5           11.4
```

Pooling gives 8.9 points. Averaging within-city gaps gives 14.5 — 63% larger. The median city is 11.4.

3/4: Computes both and reports that the within-city figure is bigger.

4/4: Answers the second half — which should a newspaper print — and defends it. The strongest case is for the within-city average, because the question “does a 1937 grade predict who lives there now” is a question *about neighbourhoods within a housing market*, and the pooled number answers a different question nobody asked.

4/4, best: Notices that the mean of city gaps weights Joliet equally with Chicago, which is its own distortion, and proposes a population-weighted mean of within-city gaps as the estimator that answers the question without either defect. **Recognising that both published numbers are wrong in different directions is the strongest available answer.**

Option B — B and C

Verified output

```
full <- names(which(tapply(tr$grade, tr$city, function(x) length(unique(x))) == 4))
s <- tr[tr$city %in% full, ]
gg <- aggregate(cbind(total, black) ~ grade + city, data = s, sum)
gg$pct <- 100 * gg$black / gg$total
c(cities_with_all_four = length(full))
```

```
#> cities_with_all_four
#>           32
```

```
round(tapply(gg$pct, gg$grade, mean), 1)
```

```
#>      A      B      C      D
#> 12.7 20.1 23.5 27.6
```

Within the 32 cities that have all four grades, the mean tract composition rises monotonically: **A 12.7, B 20.1, C 23.5, D 27.6.**

3/4: Adds B and C and observes they fall between A and D.

4/4: Says why a smooth gradient matters. If only the extremes differed, you would suspect the A and D areas were simply different kinds of places — downtown against suburb. **A monotone gradient across four grades is much harder to produce by accident**, and it is the strongest single piece of evidence in the lab.

4/4, best: Notices the steps are uneven — the A-to-B jump is 7.4 points and C-to-D is 4.1 — and asks whether the grades were equally spaced in whatever they were measuring. They were not: HOLC’s categories were qualitative, and treating A/B/C/D as an interval scale is an assumption nobody has justified. **Same lesson as DW-NOMINATE: the number is a model.**

Option C — Attack the join

No verified output; grade the reasoning.

3/4: Names centroid-in-polygon as an approximation and observes tracts get misassigned.

4/4: Produces two mechanisms that could generate the Step 6 result without any lasting HOLC effect. The best two: **(i)** D-graded areas were downtown, and downtown tracts are small, so a centroid rule assigns them differently from sprawling suburban A tracts — the join could be picking up geography, not grade; **(ii)** tract boundaries are themselves drawn partly along the same physical features (highways, rivers, rail) that the graders used, so the two are correlated by construction.

4/4, best: Says what would settle it — area-weighted allocation rather than centroids, so a tract split between grades contributes proportionally — and notes that this is precisely the published crosswalk we could not download. **A student who arrives at “the fix exists and is behind a login” has understood something true about the field.**

The measured version, if you need it. The brief reports it under “The cost of the rule, measured”, the section Figure 3 sits in: 33 of 587 tracts in Cleveland and Philadelphia carry a grade that is not the grade covering most of them, 181 overlapping tracts are dropped outright, and the median Cleveland tract has 90% of its graded land under the grade it was given. **A student who says “the rule is probably fine” can be shown a number.** What none of it settles is the direction of the resulting bias, and saying so is worth a mark.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is explaining *why* pooling destroyed the finding, not just noticing that it did.
- **A 2** is “redlining still affects neighbourhoods today,” which is the conclusion they arrived with.
- **Also a 2:** treating the reversals as errors or dropping them.
- **Reward anyone who distinguishes persistence from causation** without prompting.

Anticipated questions

- *“Did redlining cause segregation?”* — Partly, and it also recorded segregation that already existed. The honest answer is that this data cannot separate the two, and that the best research on it uses boundary discontinuities rather than the comparison we made.
- *“Why only six states?”* — Because we downloaded tract-level census data for six. The crosswalk itself covers 215 cities; extending it is a download, not a method change.
- *“Isn’t this just income?”* — Very likely entangled with it. We have race and not income; a student who wants income should be told the ACS has it and the join already exists.
- *“Are the maps online?”* — Yes, Mapping Inequality, with the original appraisal text, and the brief now draws Cleveland’s from those very polygons. **Read the D9 sheet the chapter opens with out loud.** The written justification is more persuasive than any table here, and it takes ninety seconds. Note what it says about a neighbourhood that was, on the appraiser’s own form, 0% Black: the grade is about who might arrive.

- “*How wrong is the centroid rule?*” — Measurably. In the two cities the brief draws, 33 of 587 tracts get a grade that is not the grade covering most of their land, and 181 more tracts overlap a graded area and are dropped because their centre point missed. **That is Option C’s answer arriving as a picture rather than as a worry**, and a student who has seen Figures 2 and 3 should be held to a higher standard on it.

Connections

- **The other two labs in this unit** — jury selection (a control group in the file) and policing (conditioning to escape a missing denominator). **Ask which inference the room would most want to defend, and why.**
- **Session 2 (census)** — the tract data, and the race categories.
- **Session 14 (precincts)** — the identical allocation problem: units drawn by different people for different purposes, joined by an assumption.
- **EAVS** — the first time pooling across incomparable units broke an average.